

SEONJIN NA

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SUMMARY

Seonjin Na is a **postdoctoral fellow** at **Georgia Institute of Technology**, focusing on GPU/NPU Architecture, Large Language Model (LLM) Inference Optimizations with HW/SW co-design, and Secure Architecture for GPU/NPU.

Keywords: Accelerator Architecture (GPU/NPU), Systems for ML, HW/SW Co-design, Workload Analysis, Performance Profiling

WORK EXPERIENCE

Georgia Institute of Technology: Postdoctoral Fellow

June 2023 – Present

- Advised by Prof. Hyesoon Kim and collaborated with Prof. Tushar Krishna.

Microsoft Research: Research Intern

Mar 2019 – June 2019

- Advised by Lintao Zhang and Yunxin Liu.

EDUCATION

Doctor of Philosophy, School of Computing

Feb. 2023

Korea Advanced Institute of Science and Technology (KAIST)

Advisor: Jaehyuk Huh

Master of Science, School of Computing

Feb. 2018

Korea Advanced Institute of Science and Technology (KAIST)

Advisor: Jaehyuk Huh

Bachelor of Science, Computer Science and Engineering

Feb. 2016

Sogang University

Summa Cum Laude

RESEARCH EXPERIENCE

Georgia Institute of Technology: Postdoctoral Fellow

June 2023 - Present

- **Flexible LLM Inference Serving System:** Proposed a flexible system for LLM inference that minimizes latency by leveraging both CPU and GPU hardware in offloading-based inference, using a performance estimator and extensions from HuggingFace and Intel Extension for PyTorch [2].
- **CPU LLM Inference Characterization:** Investigated the opportunity of CPU computation in offloading-based LLM inference and identified optimal CPU configurations to improve CPU LLM inference performance, using hardware performance counters and analysis with the VTune profiler [6].
- **Efficient Address Translation for MCM-GPU Systems:** Analyzed the performance impact of address translation in Multi-Chip Module (MCM) GPU systems and proposed coalesced address translation scheme with page-mapping adjustments to enhance performance [7].
- **Sampling-Based Fast GPU Simulation:** Designed a sampling-based faster GPU simulation method to accelerate cycle-level GPU simulations with statistical kernel sampling and hierarchical clustering for running various GPU workload simulations with low error rate [1], [8].
- **Efficient GPU Memory Safety Mechanism:** Addressed memory safety concerns on GPUs, such as buffer overflows, by proposing a hardware-based fine-grained GPU memory safety mechanism with minimal impact on performance [5].
- **Efficient Memory Protection for System-on-Chip System:** Investigated security mechanisms in heterogeneous SoC architectures (CPU, GPU, NPU) and developed a multi-granular memory protection scheme that dynamically adapts to diverse memory access patterns [3].
- **Contention-aware Thread Block Scheduling for Efficient GPU-SSD:** Proposed GPU thread block scheduler to mitigate SSD contention by reordering GPU thread blocks based on their memory access patterns, demonstrating average 1.68× performance improvement across diverse GPU workloads [4].

KAIST: Graduate Research Assistant

Mar 2018 – Feb 2023

- **Efficient Secure Communication for Trusted Multi-GPU Systems:** Investigated the performance overhead of securing CPU-GPU and GPU-GPU data transfers in multi-GPU systems and proposed a dynamic, batched metadata management scheme to minimize the performance impact of it, using the MGPUSim simulator [9].
- **Low-Overhead GPU Memory Protection:** Proposed an efficient memory protection scheme that leverages observed memory update behaviors. The evaluation was conducted using the GPGPU-Sim and NVBit [13].
- **Accelerating DNN Training on NPUs:** Developed dataflow transformations on NPUs to maximize data reuse of scratchpad memory (SPM) in the backward pass of DNN training, improving inter-operation data locality and reducing training latency [10].
- **Extending Trusted Execution Environments on NPUs:** Proposed a trusted NPU architecture with hardware changes and proposed efficient data protection tailored for NPUs using Scale-Sim simulator [11],[12].

PUBLICATIONS

- [1] Euijun Chung, **Seonjin Na**, Sung Ha Kang, Hyesoon Kim, **Swift and Trustworthy Large-Scale GPU Simulation with Fine-Grained Error Modeling and Hierarchical Clustering**, *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2025.
- [2] **Seonjin Na**, Geonhwa Jeong, Byunghoon Ahn, Aaron Jezghani, Jeffrey Young, Christopher J. Hughes, Tushar Krishna, Hyesoon Kim, **FlexInfer: Flexible LLM Inference with CPU Computations** , *Conference on Machine Learning and Systems (MLSys)*, 2025.
- [3] Sunho Lee, **Seonjin Na**, Jeongwon Choi, Jinwon Pyo, Jaehyuk Huh, **Unified Memory Protection with Multi-granular MAC and Integrity Tree for Heterogeneous Processors** , *International Symposium on Computer Architecture (ISCA)*, 2025.
- [4] Xueyang Liu, **Seonjin Na**, Euijun Chung, Jiashen Cao, Jing Yang, Hyesoon Kim, **Contention-Aware GPU Thread Block Scheduler for Efficient GPU-SSD** , *IEEE Computer Architecture Letters (CAL)*, 2025.
- [5] Jaewon Lee, Euijun Chung, Saurabh Singh, **Seonjin Na**, Yonghae Kim, Jaekyu Lee, Hyesoon Kim, **Let-Me-In: (Still) Employing In-pointer Bounds metadata for Fine-grained GPU Memory Safety** , *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, Mar 2025.
- [6] **Seonjin Na**, Geonhwa Jeong, Byunghoon Ahn, Jeffrey Young, Tushar Krishna, Hyesoon Kim, **Understanding Performance Implications of LLM Inference on CPUs** , *IEEE International Symposium on Workload Characterization (IISWC)*, Sep 2024.
- [7] Yuan Feng, **Seonjin Na**, Hyesoon Kim, Hyeran Jeon, **Barre Chord: Efficient Virtual Memory Translation for Multi-Chip-Module GPUs** , *International Symposium on Computer Architecture (ISCA)*, June 2024.
- [8] Euijun Chung, **Seonjin Na**, Hyesoon Kim, **Allegro: GPU Simulation Acceleration for Machine Learning Workloads** , *MLArchSys Workshop in the International Symposium on Computer Architecture (MLArchSys)*, June 2024.
- [9] **Seonjin Na**, Jungwoo Kim, Sunho Lee, Jaehyuk Huh, **Supporting Secure Multi-GPU Computing with Dynamic and Batched Metadata Management** , *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, Mar 2024.
- [10] Jungwoo Kim, **Seonjin Na**, Sanghyeon Lee, Sunho Lee, Jaehyuk Huh, **Improving Data Reuse in NPU On-chip Memory with Interleaved Gradient Order for DNN Training** , *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, Oct 2023.
- [11] Sunho Lee, **Seonjin Na**, Jungwoo Kim, Jaehyuk Huh, **Tunable Memory Protection for Secure Neural Processing Units** , *IEEE International Conference on Computer Design (ICCD)*, Oct 2022.
- [12] Sunho Lee, Jungwoo Kim, **Seonjin Na**, Jaehyuk Huh, **TNPU: Supporting Trusted Execution with Tree-less Integrity Protection for Neural Processing Unit** , *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, Mar 2022.
- [13] **Seonjin Na**, Sunho Lee, Yeonjae Kim, Jongse Park, Jaehyuk Huh, **Common Counters: Compressed Encryption Counters for Secure GPU Memory** , *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, Mar 2021.

AWARDS & HONORS

Outstanding Post-doctoral Research Award <i>College of Computing at Georgia Institute of Technology (Georgia Tech)</i>	2025
MICRO 2024 Ph.D Forum <i>2024 IEEE/ACM International Symposium on Microarchitecture (MICRO)</i>	2024
National Scholarship <i>Korea Advanced Institute of Science and Technology (KAIST)</i>	2016-2023
Summa Cum Laude <i>Sogang University</i>	2016
Gold Prize <i>ACM-ICPC Asia Daejeon Regional Contest 4th place</i>	2015
Honorable Mention <i>ACM-ICPC Asia Daejeon Regional Contest 13th place</i>	2013
Academic Scholarship, 8 semesters <i>Sogang University</i>	2012-2015

PATENTS

- [P1] Jaehyuk Huh, Sunho Lee, **Seonjin Na**, **Apparatus and Method for Providing Secure Execution Environment for NPU** , Patent US 12045337, United States, 2024
- [P2] Jaehyuk Huh, **Seonjin Na**, Jungwoo Kim, Sunho Lee, **Dynamic One-time Pad Table Management for Secure Multi-GPU Communication**, Patent KR 1020230055347, South Korea, 2023
- [P3] Jaehyuk Huh, Jungwoo Kim, **Seonjin Na**, Sanghyeon Lee, Sunho Lee, **Improving the Utilization of NPU On-chip Memory with Computation Rearrangement for DNN Training**, Patent KR 1020230055346, South Korea, 2022
- [P4] Jaehyuk Huh, Sunho Lee, **Seonjin Na**, **Hardware-based Security Architecture for Trusted Neural Processing Unit**, Patent KR 1020220055977, South Korea, 2022
- [P5] Jaehyuk Huh, **Seonjin Na**, Sunho Lee, Yeonjae Kim, and Jongse Park, **Efficient Encryption Method and Apparatus for Hardware-based Secure GPU Memory**, Patent KR 1023652630000, South Korea, 2022

ACADEMIC SERVICES

Technical Program Committee

- Architectural Support for Programming Languages and Operating Systems (ASPLOS) 2026
- General Purpose Processing on Graphics Processing Units (GPGPU) 2025
- International Parallel & Distributed Processing Symposium (IPDPS) 2025
- International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC) 2024

Journal Reviewer

- ACM Transactions on Computer Systems (TOCS) 2025
- ACM Transactions on Architecture and Code Optimization (TACO) 2025
- IEEE Computer Architecture Letter (CAL) 2025
- ACM Transactions on Computer Systems (TOCS) 2024
- ACM Transactions on Architecture and Code Optimization (TACO) 2024 x 4
- IEEE Transactions on Dependable and Secure Computing (TDSC) 2023
- IEEE Computer Architecture Letter (CAL) 2023

Workshop/Tutorial Chair

- IEEE International Symposium on Workload Characterization (IISWC) 2025

Travel Grant Chair

- Architectural Support for Programming Languages and Operating Systems (ASPLOS) 2025

Web Chair

- IEEE Computer Society TCuARCH
- Vortex Workshop and Tutorial at MICRO 2024

Artifact Evaluation Committee

- IEEE/ACM International Symposium on Microarchitecture (MICRO) 2025
- IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) 2025
- ACM European Conference on Computer Systems (EuroSys) 2025
- Architectural Support for Programming Languages and Operating Systems (ASPLOS) 2025
- IEEE/ACM International Symposium on Microarchitecture (MICRO) 2024
- USENIX Symposium on Operating Systems Design and Implementation (OSDI) 2024
- USENIX Annual Technical Conference (ATC) 2024
- International Symposium on Computer Architecture (ISCA) 2024

TECHNICAL SKILLS

Data Analysis and Profiling Tool: Pandas, Matplotlib, Intel VTune, Linux Perf, NVIDIA Nsight Profiler

Machine Learning Framework: Pytorch, vLLM, DeepSpeed, HuggingFace, Intel Extension for Pytorch

Architectural Simulator: GPGPU-Sim, MGPUSim, ScaleSim, Gem5, ChampSim, Macsim

Programming Language and Others: Python, C, C++, Go, CUDA, NVBit